Politecnico di Milano
School of Industrial and Information Engineering

Nonparametric Statistics

Acadaemic Year 2026/2027

Teachers

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Teaching Assistant

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Standard Schedule

Tuesday, 10:15-13:15, Classroom 9.0.3.
Thursday, 11:15-13:15, Classroom 26.0.4
Friday, 13:15-15:15, Classroom 3.1.9

Communications about possible variations to the course schedule will be given through Webeep Channel.

During the lessons and lab sessions, several topics and exercises will be proposed through the use of R software.

Program

The course aims to introduce students to some basic and advanced topics of nonparametric statistics, and to provide students with statistical and algorithmic tools for the analysis of complex data (i.e., data for which the identification of standard and classical probabilistic models is hard or not possible) through a nonparametric approach.

The main topics of the course will be: descriptive and inference for multivariate data in a nonparametric setting, nonparametric regression, nonparametric forecasting. Specifically:

- **Part I**: Nonparametric Exploration.
  - Depth measures
  - *Depth measures for functional data [only for the 8 CFU version]*
- **Part II**: Nonparametric Inference.
  - Rank Tests
  - Permutational Inference (Tests and Confidence Intervals for means, linear regression, and anova)
  - Bootstrap Inference (Tests and Confidence Intervals for means, linear regression, and anova)
  - *Permutational Inference for functional data [only for the 8 CFU version]*
- **Part III**: Nonparametric Regression.
  - Nonparametric Regression: Kernel methods (univariate and Multivariate); Splines; Semiparametric regression; Spline with differential penalization
  - Quantile Regression
  - Nonparametric estimation in Survival Analysis: Kaplan-Meier and Nelson-Aalen estimators, Semiparametric Cox model
- **Part IV**: Nonparametric Forecasting.
  - Conformal prediction
  - *Conformal prediction for functional data [only for the 8 CFU version]*

Exam and final evaluation

The course assessment will consist of a composite individual evaluation (homework + oral discussion) and a team project for both the 5CFU and 8CFU version of the course.

The composite individual evaluation will consist of two parts: firstly, a homework where each student will be required to provide the solution of an execise in terms of code necessary for getting the solution; then, an oral discussion where the student will be required to solve an exercise and to answer some theoretical questions. Failing the first part will prevent to access the second part of the composite evaluation. The composite individual evaluation will be scored on a scale from 0 to 30, the maximum evaluation being 32/30. It will be passed upon obtaining a score larger or equal to 18/30. The exam evaluation will account for the degree of clarity of the exposition and for the correctness of methods and computations. For each question students will be asked to report both the method used and the numerical solution obtained, and to critically discuss the latter one.

The team project will consist of an analysis of a real dataset, to be conducted in teams of 2 to 4 students, using the models and methods introduced in the course. The team projects will be presented during and at the end of the course in an open workshop that will take place after the end of the semester. For the groups whose students are not willing to give the written exam during the winter semester a second date will be available during the summer session. Each team will receive an evaluation in a scale from 0 to 30. More details about the expected project development will be given in a dedicated lesson.

The final individual grade will be obtained as a weighted average of the grades obtained by the student in the two parts of the evaluation: **2/3 grade of the composite exam + 1/3 grade of the project**.

During the exam, the students will have to:

- Demonstrate the degree of knowledge and comprehension of the key aspects of the course, presenting the used methodologies in a clear and exhaustive way;

- Demonstrate their ability to apply the learned notions to solve exercises and real problems, on any of the topics covered in the course.

Requirements

Strongly suggested: Metodi e Modelli per l'Inferenza Statistica (BSc in Mathematical Engineering), Applied Statistics (MSc in Mathematical Engineering).

Bibliography

References to books and papers are provided on the official page of the course. Further references to specific topics will be provided by the teachers during the course.